Combination Numbers

- Pascal's Rule: $C_n^k = C_{n-1}^k + C_{n-1}^{k-1}$.
- Absorption Identity: $kC_n^k = nC_{n-1}^{k-1}$.
- Trinomial Revision: $C_n^m C_m^k = C_n^k C_{n-k}^{m-k}$.
- Row Sum: $\sum_{k=0}^n C_n^k = 2^n$.
- Even-Odd Sums: $\sum_{k\ge 0} C_n^{2k} = \sum_{k\ge 0} C_n^{2k+1} = 2^{n-1}$.
- Weighted Sums:
  - $\sum_{k=0}^n kC_n^k = n2^{n-1}$.
  - $\sum_{k=0}^n k^2 C_n^k = n(n+1)2^{n-2}$.
  - $\sum_{k=0}^n k^3 C_n^k = n^2(n+3)2^{n-3}$.
  - $\sum_{k=0}^m (-1)^k C_n^k = (-1)^m C_{n-1}^m$.
- Hockey-Stick Identity: $\sum_{i=k}^n C_i^k = C_{n+1}^{k+1}$.
- Diagonal Sum: $\sum_{k=0}^m C_{n+k}^k = C_{n+m+1}^m$.
- Vandermonde's Identity: $\sum_{k=0}^r C_n^k C_m^{r-k} = C_{n+m}^r$.

Probability Space

- **Sample Space** Ω: the set of all possible outcomes.
- **σ-algebra**: the collection of events.
  - $\emptyset \in \mathcal{F}$,
  - If $A \in \mathcal{F}$, then $A^{\complement} \in \mathcal{F}$,
  - If $A_i \in \mathcal{F}$, then $\bigcup_i A_i \in \mathcal{F}$.
- **Borel field**: all intervals and σ-algebra generated from them.
- **Probability Measure P**: a function $\mathbf{P}: \mathcal{F} \to [0,1]$.
  - Nonnegativity: $\mathbf{P}(A) \ge 0, \forall A \in \mathcal{F}$,
  - Normalization: $\mathbf{P}(\emptyset) = 0, \mathbf{P}(\Omega) = 1$,
  - Countable additivity: If $A_i \in \mathcal{F}$ are *disjoint*, $\mathbf{P}\left(\bigcup_i A_i\right) = \sum_i \mathbf{P}(A_i)$.

Conditioning and Independence

Conditioning

- **Definition**: For $\mathbf{P}(B) \ne 0$, $\mathbf{P}(A|B) = \mathbf{P}(A \cap B)/\mathbf{P}(B)$.
  - Nonnegativity: $\mathbf{P}(A|B) \ge 0, \forall A \in \mathcal{F}$,
  - Normalization: $\mathbf{P}(\Omega|B) = 1$,
  - Countable additivity: If $A_i \in \mathcal{F}$ are *disjoint*, $\mathbf{P}\left(\bigcup_i A_i | B\right) = \sum_i \mathbf{P}(A_i|B)$.
- **Multiplication Rule**: $\mathbf{P}(A \cap B \cap C) = \mathbf{P}(A) \cdot \mathbf{P}(B|A) \cdot \mathbf{P}(C|A \cap B)$.
- **Total Probability Theorem**: For a partition of sample space $A_1, A_2, \ldots, A_n$, assuming $\mathbf{P}(A_i) > 0$ for all i,

$$\mathbf{P}(B) = \sum_{i=1}^n \mathbf{P}(A_i)\mathbf{P}(B|A_i).$$

- **Bayes's Rule**: $\mathbf{P}(A_i|B) = \dfrac{\mathbf{P}(A_i)\mathbf{P}(B|A_i)}{\sum_{j=1}^n \mathbf{P}(A_j)\mathbf{P}(B|A_j)}$.

Independence

- **Definition**: Event A and B are called **independent** if $\mathbf{P}(A \cap B) = \mathbf{P}(A)\mathbf{P}(B)$.
- For $\mathbf{P}(B) > 0$, $\mathbf{P}(A|B) = \mathbf{P}(A)$.
- **Conditional Independence**: $\mathbf{P}(A \cap B|C) = \mathbf{P}(A|C)\mathbf{P}(B|C)$.
- **Collective Independence**: Events $A_1, A_2, \ldots, A_n$ are called independent if

$$\mathbf{P}(A_i \cap A_j \cap \cdots \cap A_q) = \mathbf{P}(A_i)\mathbf{P}(A_j)\ldots\mathbf{P}(A_q),$$

  for any distinct indices $i, j, \ldots, q$ chosen from $\{1, 2, \ldots, n\}$.
- **Inclusion-Exclusion Principle**: $\mathbf{P}\left(\bigcup_{i=1}^n A_i\right) = \sum \mathbf{P}(A_i) - \sum_{i<j} \mathbf{P}(A_i \cap A_j) + \cdots + (-1)^{n-1}\mathbf{P}(A_1 \cap \cdots \cap A_n)$.
  - **Hat Problem (Derangement)**:
    - * Derangement Number:
      $D_n = n! \sum_{i=0}^n (-1)^i / i!$.
      $D_n = nD_{n-1} + (-1)^n$.
      $\lim_{n\to\infty} D_n / n! = 1/\mathrm{e}$.
    - * Prob. of at least one match is $\approx 1 - 1/\mathrm{e}$ as $n \to \infty$.
  - **Symmetric Difference**: $\mathbf{P}(A \triangle B) = \mathbf{P}(A) + \mathbf{P}(B) - 2\mathbf{P}(A \cap B)$.

Discrete Random Variable

- **Random Variable**: Given a probability space $(\Omega, \mathcal{F}, \mathbf{P})$, a **random variable** is a *function* $X: \Omega \to \mathbb{R}$ such that $X^{-1}(A) \in \mathcal{F}$ for each $A \in \mathcal{B}(\mathbb{R})$.
- **PMF**:
  $p_X(x) = \mathbf{P}(X = x)$.
- **Derived PMF**: $Y = g(X) \Rightarrow p_Y(y) = \sum_{x:g(x)=y} p_X(x)$.
- **Expectation**:
  $\mathbf{E}[X] = \sum_i x_i p_X(x_i)$.
- **Variance**:
  $\mathrm{var}[X] = \mathbf{E}\left[(X - \mathbf{E}[X])^2\right] = \mathbf{E}[X^2] - (\mathbf{E}[X])^2$.
- $\mathbf{E}[\alpha X + \beta] = \alpha \mathbf{E}[X] + \beta$, $\mathbf{E}[X + Y] = \mathbf{E}[X] + \mathbf{E}[Y]$.
- $\mathrm{var}[\alpha X + \beta] = \alpha^2 \mathrm{var}[X]$.
- **Geometric RV**:
  - $p_X(k) = (1-p)^{k-1}p$,
  - $\mathbf{E}[X] = 1/p$,
  - $\mathbf{E}[X^2] = (2-p)/p^2$,
  - $\mathrm{var}[X] = (1-p)/p^2$.
  - **Memorylessness**:
    $p_X(k|X > n) = (1-p)^{k-n-1}p$; $\mathbf{E}[X|X > n] = 1/p + n$.
- **Bernoulli RV**:
  - $p_X(k) = \begin{cases} p, & k = 1, \\ 1-p, & k = 0, \end{cases}$,
  - $\mathbf{E}[X] = p$,
  - $\mathbf{E}[X^2] = p$,
  - $\mathrm{var}[X] = p(1-p)$.
- **Discrete Uniform RV**:
  - $p_X(k) = 1/(b-a+1)$, if $k = a, a+1, \ldots, b$,
  - $\mathbf{E}[X] = (a+b)/2$,
  - $\mathrm{var}[X] = (b-a)(b-a+2)/12$.
- **Binomial RV**:
  - $p_X(k) = C_n^k p^k (1-p)^{n-k}$,
  - $\mathbf{E}[X] = np$,
  - $\mathbf{E}[X^2] = np(1-p+np)$,
  - $\mathrm{var}[X] = np(1-p)$.
- **Poisson RV**: $\lambda = np$ as $n \to \infty$ in binomial RV.
  - $p_X(k) = \mathrm{e}^{-\lambda}\lambda^k / k!$, $k = 0, 1, \ldots$,
  - $\mathbf{E}[X] = \lambda$,
  - $\mathbf{E}[X^2] = \lambda + \lambda^2$,
  - $\mathrm{var}[X] = \lambda$,
  - Independent sum: $X_1 + X_2 \sim \mathrm{Poisson}(\lambda_1 + \lambda_2)$.

Multiple Discrete RV

- **Conditioning**:
  - **Conditional PMF**:
    $p_{X|A}(x) = \mathbf{P}(\{X = x\} \cap A)/\mathbf{P}(A)$, $p_{X|Y}(x|y) = p_{X,Y}(x,y)/p_Y(y)$.
  - **Conditional Expectation**:
    $\mathbf{E}[X|A] = \sum_x x p_{X|A}(x)$.
  - **Total Expectation Theorem**:
    $\mathbf{E}[X] = \mathbf{P}(A_1)\mathbf{E}[X|A_1] + \cdots + \mathbf{P}(A_n)\mathbf{E}[X|A_n]$.
  - **Marginal PMF**:
    $p_{X,Y}(x,y) = \sum_z p_{X,Y,Z}(x,y,z)$; $p_X(x) = \sum_y p_{X,Y}(x,y)$.
- **Independence**:
  $\forall x, y$: $p_{X,Y}(x,y) = p_X(x)p_Y(y)$.
- For Independent RVs:
  - $f(X)$ and $g(Y)$ are also independent.
  - $\mathbf{E}[XY] = \mathbf{E}[X]\mathbf{E}[Y]$.
  - $\mathrm{var}(X+Y) = \mathrm{var}(X) + \mathrm{var}(Y)$, $\mathrm{var}(X-Y) = \mathrm{var}(X) + \mathrm{var}(Y)$.
  - **Convolution**: $W = X + Y \Rightarrow p_W(w) = \sum_x p_X(x)p_Y(w-x)$.
- **Conditional Independence**:
  $\forall x, y$: $p_{X,Y}(x,y|A) = p_X(x|A)p_Y(y|A)$.

Continuous RV

- **PDF**:
  $\mathbf{P}(a \le X \le b) = \int_a^b f_X(x)\,\mathrm{d}x$;
  $\mathbf{P}(x \le X \le x + \delta) \approx f_X(x)\delta$.
- **Expectation**:
  $\mathbf{E}[X] = \int_{-\infty}^{+\infty} x f_X(x)\,\mathrm{d}x$.
- **Variance**:
  $\mathrm{var}(X) = \int_{-\infty}^{+\infty} (x - \mathbf{E}[X])^2 f_X(x)\,\mathrm{d}x = \mathbf{E}[X^2] - (\mathbf{E}[X])^2$.
- **Uniform RV**:
  - $f_X(x) = 1/(b-a)$, $a \le x \le b$,
  - $\mathbf{E}[X] = (a+b)/2$,
  - $\mathbf{E}[X^2] = (a^2 + ab + b^2)/3$,
  - $\mathrm{var}[X] = (b-a)^2/12$.
- **Exponential RV**:
  - $f_X(x) = \lambda \mathrm{e}^{-\lambda x}$, $x \ge 0$,
  - $F_X(x) = 1 - \mathrm{e}^{-\lambda x}$, $x \ge 0$.
  - $\mathbf{E}[X] = 1/\lambda$,
  - $\mathbf{E}[X^2] = 2/\lambda^2$,
  - $\mathrm{var}[X] = 1/\lambda^2$.
  - $\mathbf{P}(X_1 < X_2) = \lambda_1/(\lambda_1 + \lambda_2)$.
  - Memorylessness:
    $f_X(x|X > a) = \lambda \mathrm{e}^{-\lambda(x-a)}$, $\mathbf{P}(X \ge c + x | X \ge c) = \mathbf{P}(X \ge x) = \mathrm{e}^{-\lambda x}$;
    $\mathbf{E}[X|X > c] = 1/\lambda + c$.
- **Gaussian RV** ($X \sim N(\mu, \sigma^2)$):
  - $$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \mathrm{e}^{-(x-\mu)^2/(2\sigma^2)},$$
  - $F_\Lambda(\lambda) = \Phi(\lambda)$, $\Lambda \sim N(0,1)$.
  - $\mathbf{E}[X] = \mu$,
  - $\mathbf{E}[X^2] = \mu^2 + \sigma^2$,
  - $\mathbf{E}[X^4] = 3\sigma^4 + 6\sigma^2\mu^2 + \mu^4$,
  - $\mathrm{var}[X] = \sigma^2$.
  - $Y = \alpha X + \beta \sim N(\alpha\mu + \beta, \alpha^2\sigma^2)$.
- **CDF**:
  - $F_X(x) = \int_{-\infty}^{x+} f_X(t)\,\mathrm{d}t$.
  - $F_X(x)$ discontinuous $\Rightarrow X$ has discrete part.
  - $\mathbf{P}(a < X \le b) = F_X(b) - F_X(a)$.
  - $\mathbf{E}[X] = \int_0^{+\infty}(1 - F_X(x))\,\mathrm{d}x - \int_{-\infty}^0 F_X(x)\,\mathrm{d}x$.
- **Derived PDF**:
  - **CDF Method**: find $F_Y(y) = \mathbf{P}(g(X) \le y) \Rightarrow f_Y(y) = F_Y'(y)$.
  - **Linear**: $Y = aX + b \Rightarrow f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$.
  - **Monotone**: $y = g(x) \Leftrightarrow x = h(y) \Rightarrow f_Y(y) = f_X(h(y))|h'(y)|$.
- **Min and Max of i.i.d. RVs** $(X_1, \ldots, X_n)$:
  - **Max**: $Z = \max(X_i) \implies F_Z(z) = [F_X(z)]^n \implies f_Z(z) = n[F_X(z)]^{n-1}f_X(z)$.
  - **Min**: $W = \min(X_i) \implies F_W(w) = 1 - [1 - F_X(w)]^n$.
  - *Shortcut*: If $X_i \sim \mathrm{Exp}(\lambda_i)$, then $\min(X_1, \ldots, X_n) \sim \mathrm{Exp}(\sum \lambda_i)$.
- **2D Uniform & Geometric Probability**:
  - $f_{X,Y}(x,y) = 1/\mathrm{Area}(S)$ for $(x,y) \in S$.
  - **Broken Stick / Triangle Inequality**: Break a line into 3 pieces. They form a triangle iff. all pieces are shorter than 1/2. Prob. is 1/4.

Multiple Continuous RV

Conditioning:
- **Conditional PDF**: $f_{X|Y}(x|y) = f_{X,Y}(x,y)/f_Y(y)$.
- **Conditional Expectation**: $\mathbf{E}[X|Y = y] = \int_{-\infty}^{+\infty} x f_{X|Y}(x|y)\,\mathrm{d}x$.
- **Total Expectation Theorem**: $\mathbf{E}[X] = \int_{-\infty}^{+\infty} \mathbf{E}[X|Y = y] f_Y(y)\,\mathrm{d}y$.
- **Marginal PDF**: $f_X(x) = \int_{-\infty}^{+\infty} f_{X,Y}(x,y)\,\mathrm{d}y$.

Independence:
- $\forall x, y$: $f_{X,Y}(x,y) = f_X(x)f_Y(y)$.
- For independent RVs:
  - $f(X)$ and $g(Y)$ are also independent,
  - $\mathbf{E}[XY] = \mathbf{E}[X]\mathbf{E}[Y]$,
  - $\mathrm{var}(X+Y) = \mathrm{var}(X) + \mathrm{var}(Y)$.
  - **Convolution**: If X and Y are independent and $Z = X + Y$, $f_Z(z) = (f_X * f_Y)(z) = \int_{-\infty}^{+\infty} f_X(x)f_Y(z-x)\,\mathrm{d}x$.
  - $X \sim N(\mu_X, \sigma_X^2)$, $Y \sim N(\mu_Y, \sigma_Y^2) \Rightarrow X + Y \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2)$.

Joint / Multiple CDF: $F_{X,Y}(x,y) = \mathbf{P}(X \le x, Y \le y)$.
- **Marginals**: $F_X(x) = F_{X,Y}(x, \infty)$, $F_Y(y) = F_{X,Y}(\infty, y)$.
- **From/To PDF**:
  $F_{X,Y}(x,y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(u,v)\,\mathrm{d}v\,\mathrm{d}u$
  $\implies f_{X,Y}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x,y)$.
- **Independence**: $X \perp Y \iff F_{X,Y}(x,y) = F_X(x)F_Y(y)$.

Conditional CDF:
- **Given Event** A: $F_{X|A}(x) = \mathbf{P}(X \le x \mid A) = \frac{\mathbf{P}(\{X \le x\} \cap A)}{\mathbf{P}(A)}$.
- **Given** $Y = y$: $F_{X|Y}(x|y) = \int_{-\infty}^x f_{X|Y}(t|y)\,\mathrm{d}t$.
- **Derivative**: $f_{X|Y}(x|y) = \frac{\partial}{\partial x} F_{X|Y}(x|y)$.
- **Total Prob**: $F_X(x) = \sum \mathbf{P}(A_i)F_{X|A_i}(x)$.

Continuous Bayes' Rule:

$$f_{X|Y}(x|y) = \frac{f_{Y|X}(y|x)f_X(x)}{f_Y(y)} = \frac{f_{Y|X}(y|x)f_X(x)}{\int_{-\infty}^{\infty} f_{Y|X}(y|t)f_X(t)\,dt},$$

$$\mathbf{P}(A|Y = y) = \frac{\mathbf{P}(A)f_{Y|A}(y)}{f_Y(y)}. \text{(Mixed Bayes)}$$

Covariance and Correlation

- **Covariance**: $\mathrm{cov}(X,Y) = \mathbf{E}[(X - \mathbf{E}[X])(Y - \mathbf{E}[Y])] = \mathbf{E}[XY] - \mathbf{E}[X]\mathbf{E}[Y]$.
- **Uncorrelated**: $\mathrm{cov}(X,Y) = 0$; independent $\Rightarrow \mathrm{cov}(X,Y) = 0$.
- $\mathrm{cov}(X,X) = \mathrm{var}(X)$; $\mathrm{cov}(X, aY + b) = a\,\mathrm{cov}(X,Y)$; $\mathrm{cov}(X, Y + Z) = \mathrm{cov}(X,Y) + \mathrm{cov}(X,Z)$.
- $\mathrm{var}(X+Y) = \mathrm{var}(X) + \mathrm{var}(Y) + 2\,\mathrm{cov}(X,Y)$.
- $\mathrm{var}(\sum_i X_i) = \sum_i \mathrm{var}(X_i) + \sum_{i \ne j} \mathrm{cov}(X_i, X_j)$.
- **Correlation**: $\rho(X,Y) = \dfrac{\mathrm{cov}(X,Y)}{\sqrt{\mathrm{var}(X)\,\mathrm{var}(Y)}}$, $-1 \le \rho \le 1$.
- $|\rho| = 1 \Leftrightarrow X - \mathbf{E}[X] = c(Y - \mathbf{E}[Y])$.

Conditional Expectation

- **Iterated Expectation**: $\mathbf{E}[X] = \mathbf{E}[\mathbf{E}[X|Y]]$.
  - Y discrete: $\mathbf{E}[\mathbf{E}[X|Y]] = \sum_y \mathbf{E}[X|Y = y]p_Y(y)$.
  - Y continuous: $\mathbf{E}[\mathbf{E}[X|Y]] = \int \mathbf{E}[X|Y = y]f_Y(y)\,\mathrm{d}y$.
- **Estimator**: $\hat{X} = \mathbf{E}[X|Y]$, $\tilde{X} = X - \hat{X}$.
- $\mathbf{E}[\tilde{X}|Y] = 0 \Rightarrow \mathbf{E}[\tilde{X}] = 0$; $\mathrm{cov}(\hat{X}, \tilde{X}) = 0$.
- **Conditional Variance**: $\mathrm{var}(X|Y) = \mathbf{E}[(X - \mathbf{E}[X|Y])^2|Y]$.
- **Total Variance**: $\mathrm{var}(X) = \mathrm{var}(\mathbf{E}[X|Y]) + \mathbf{E}[\mathrm{var}(X|Y)]$.

Transforms

- **Definition**: $\mathscr{M}_X(s) = \mathbf{E}[e^{sX}]$;
  discrete $\sum_x p_X(x)e^{sx}$; continuous $\int f_X(x)e^{sx}\,dx$.
- **Recover Law**: MGF exists near 0 $\Rightarrow$ unique distribution.
  - Discrete: expand $\mathscr{M}_X(s) = \sum_x p_X(x)e^{sx}$; coeff. of e^{sx} is $p_X(x)$; or $p_X(x) = \mathcal{Z}^{-1}\{\mathscr{M}_X(-\ln z)\}$.
  - Continuous: $f_X(x) = \mathcal{L}_b^{-1}\{\mathscr{M}_X(-s)\}$; or $f_X(x) = \mathcal{F}^{-1}\{\mathscr{M}_X(it)\} = \frac{1}{2\pi}\int e^{-itx}\mathscr{M}_X(it)\,dt$.
- $\mathscr{M}_X(0) = 1$; $\mathbf{E}[X^n] = \left.\frac{d^n\mathscr{M}_X(s)}{ds^n}\right|_{s=0}$.
- $Y = aX + b \Rightarrow \mathscr{M}_Y(s) = e^{bs}\mathscr{M}_X(as)$.
- Independent $Z = \sum_{i=1}^n X_i \Rightarrow \mathscr{M}_Z(s) = \prod_i \mathscr{M}_{X_i}(s)$.
- **Mixture**: $f_Y = \sum_i p_i f_{X_i} \Rightarrow \mathscr{M}_Y(s) = \sum_i p_i \mathscr{M}_{X_i}(s)$.
- **Random Sum**: $Y = X_1 + \cdots + X_N$, i.i.d. $X_i \perp N$:
  - $\mathbf{E}[Y] = \mathbf{E}[N]\mathbf{E}[X]$,
  - $\mathrm{var}(Y) = \mathbf{E}[N]\mathrm{var}(X) + (\mathbf{E}[X])^2\mathrm{var}(N)$,
  - $\mathscr{M}_Y(s) = \mathscr{M}_N(\ln\mathscr{M}_X(s))$.
- **Common** $\mathscr{M}_X(s)$:
  - Bernoulli: $1 - p + pe^s$; Binomial: $(1 - p + pe^s)^n$,
  - Geometric: $\frac{pe^s}{1-(1-p)e^s}$; Poisson: $\exp(\lambda(e^s - 1))$,
  - Discrete uniform $\{a, \dots, b\}$: $\frac{e^{as}(1-e^{(b-a+1)s})}{(b-a+1)(1-e^s)}$,
  - Uniform $[a, b]$: $\frac{e^{bs}-e^{as}}{(b-a)s}$; Exponential: $\frac{\lambda}{\lambda - s}$,
  - Gaussian $N(\mu, \sigma^2)$: $\exp(\mu s + \sigma^2 s^2/2)$.
  - Pascal: $\left(\frac{pe^s}{1-(1-p)e^s}\right)^k$; NegBin: $\left(\frac{p}{1-(1-p)e^s}\right)^k$.

Probability Inequalities

- **Markov**: $X \geq 0, a > 0 \Rightarrow \mathbf{P}(X \geq a) \leq \mathbf{E}[X]/a$.
  - "=" iff $X \in \{0, a\}$.
- **Chebyshev**: $\mathbf{P}(|X - \mathbf{E}[X]| \geq c) \leq \mathrm{var}(X)/c^2$.
  - "=" iff $X \in \{\mu - c, \mu, \mu + c\}$, $\mathbf{P}(X = \mu \pm c) = \sigma^2/(2c^2)$.

- **Variance Bound**: $X \in [a, b] \Rightarrow \mathrm{var}(X) \leq (b-a)^2/4$.
  - "=" iff $\mathbf{P}(X = a) = \mathbf{P}(X = b) = 1/2$.
- **Chernoff**:
  - Right-tail: $\mathbf{P}(X \geq a) \leq \inf_{s\geq 0} e^{-sa}\mathscr{M}_X(s) = e^{-\sup_{s\geq 0}(sa - \ln\mathscr{M}_X(s))}$,
  - Left-tail: $\mathbf{P}(X \leq a) \leq \inf_{s\leq 0} e^{-sa}\mathscr{M}_X(s) = e^{-\sup_{s\leq 0}(sa - \ln\mathscr{M}_X(s))}$.

LLN and CLT

- **WLLN**: $M_n = \frac{1}{n}\sum_{i=1}^n X_i$, i.i.d., $\mathbf{E}[X_i] = \mu$, $\mathrm{var}(X_i) = \sigma^2$.
  - $\mathbf{E}[M_n] = \mu$, $\mathrm{var}(M_n) = \sigma^2/n$,
  - $\mathbf{P}(|M_n - \mu| \geq \epsilon) \leq \sigma^2/(n\epsilon^2) \to 0$,
  - $M_n \xrightarrow{\mathbf{P}} \mu$.
- **Convergence**:
  - **In Probability**: $X_n \xrightarrow{\mathbf{P}} X \Leftrightarrow \mathbf{P}(|X_n - X| \geq \epsilon) \to 0, \forall\epsilon > 0$,
  - **In Distribution**: $X_n \xrightarrow{D} X \Leftrightarrow F_{X_n}(x) \to F_X(x)$ at continuous x,
  - **Almost Surely**: $X_n \xrightarrow{\text{a.s.}} X \Leftrightarrow \mathbf{P}(\lim_n X_n = X) = 1$,
    - "Almost Surely": For almost all sample paths ω.
  - r**-th Mean**: $X_n \xrightarrow{r} X \Leftrightarrow \mathbf{E}[|X_n - X|^r] \to 0$,
  - **Relations**: a.s. $\Rightarrow \mathbf{P} \Rightarrow D$; r-mean $\Rightarrow \mathbf{P} \Rightarrow D$; $r \geq s \Rightarrow L^r \Rightarrow L^s$.
- **CLT**: $S_n = \sum_{i=1}^n X_i$, i.i.d., $\mathbf{E}[X_i] = \mu$, $\mathrm{var}(X_i) = \sigma^2$.
- $Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{D} N(0, 1)$; $\mathbf{P}(Z_n \leq z) \to \Phi(z)$.
- **Transform**: $\mathscr{M}_{Z_n}(s) = e^{-s\mu\sqrt{n}/\sigma}\left(\mathscr{M}_{X_1}\left(\frac{s}{\sigma\sqrt{n}}\right)\right)^n \to e^{s^2/2}$.
- **Normal Approx.**: $S_n \approx N(n\mu, n\sigma^2)$; $\mathbf{P}(S_n \leq c) \approx \Phi\left(\frac{c - n\mu}{\sigma\sqrt{n}}\right)$.
- **Binomial Approx.**: $X_i \sim$ Bernoulli(p), $S_n = \sum_i X_i \sim$ Bin(n, p), $q = 1 - p$.
  - $\frac{S_n - np}{\sqrt{npq}} \approx N(0, 1)$,
  - $\mathbf{P}(k \leq S_n \leq l) \approx \Phi\left(\frac{l + 0.5 - np}{\sqrt{npq}}\right) - \Phi\left(\frac{k - 0.5 - np}{\sqrt{npq}}\right)$.
- **SLLN**: $M_n = S_n/n \Rightarrow \mathbf{P}(\lim_{n\to\infty} M_n = \mu) = 1$, i.e. $M_n \xrightarrow{\text{a.s.}} \mu$.

Event Limits and Borel-Cantelli

- **Set Limits**:
  - $\limsup A_n = \bigcap_{k\geq 1}\bigcup_{n\geq k} A_n = \{A_n \text{ i.o.}\}$;
    - "infinite often"$= \forall N \geq 0, \exists n \geq N, A_n$ happens.
  - $\liminf A_n = \bigcup_{k\geq 1}\bigcap_{n\geq k} A_n = \{A_n \text{ eventually}\}$.
    - "eventually"$= \exists N \geq 0, \forall n \geq N, A_n$ happens.
  - $\liminf A_n \subseteq \limsup A_n$; "=" $\Rightarrow \lim A_n$ exists.
- **Monotone Convergence Theorem**: $0 \leq X_n \uparrow X \Rightarrow \mathbf{E}[X_n] \uparrow \mathbf{E}[X]$; events $A_n \uparrow A \Rightarrow \mathbf{P}(A_n) \uparrow \mathbf{P}(A)$.
  - Let X_n = the sum of an RV seq. $\{Y_i\}$, $Y_i \geq 0$, $X = \lim_n X_n$, $\mathbf{E}[X] < \infty \Rightarrow \mathbf{P}(X = \infty) = 0 \Rightarrow X$ converge a.s..
  - $X = \sum_{i=1}^\infty Y_i$ converges a.s. $\Rightarrow \{Y_i\} \xrightarrow{\text{a.s.}} 0$.
- **Borel-Cantelli I**: $\sum_n \mathbf{P}(A_n) < \infty \Rightarrow \mathbf{P}(A_n \text{ i.o.}) = 0$;
- **Borel-Cantelli II**: independent A_n, $\sum_n \mathbf{P}(A_n) = \infty \Rightarrow \mathbf{P}(A_n \text{ i.o.}) = 1$.
- **a.s. Criterion**: $X_n \xrightarrow{\text{a.s.}} X \Leftrightarrow \forall\epsilon > 0$, $\mathbf{P}(|X_n - X| > \epsilon \text{ i.o.}) = 0$.
  - Equiv.: $\lim_{n\to\infty}\mathbf{P}(\sup_{k\geq n}|X_k - X| > \epsilon) = 0$, $\forall\epsilon > 0$,
  - Enough: $\forall m \geq 1$, $\sum_n \mathbf{P}(|X_n - X| > 1/m) < \infty$.
- **Subsequence**: $X_n \xrightarrow{\mathbf{P}} X \Rightarrow \exists n_k : X_{n_k} \xrightarrow{\text{a.s.}} X$;
  - proof: choose $\mathbf{P}(|X_{n_k} - X| > 1/k) < 2^{-k}$.
- **Kolmogorov Ineq.**: independent RVs X_i, $\mathbf{E}[X_i] = 0$, $\mathrm{var}(X_i) < \infty$, $S_k = \sum_{i=1}^k X_i \Rightarrow \mathbf{P}(\max_{k\leq n}|S_k| \geq a) \leq \mathrm{var}(S_n)/a^2 = \sum_{i=1}^n \mathrm{var}(X_i)/a^2$.
- **Cantelli SLLN**: i.i.d., $\mathbf{E}[X^4] < \infty \Rightarrow \mathbf{E}[(\sum_{i=1}^n(X_i - \mu))^4] = O(n^2)$;
  Borel-Cantelli I $\Rightarrow S_n/n \xrightarrow{\text{a.s.}} \mu$.
- $\limsup A_n$ is event; $\limsup \mathbf{P}(A_n)$ is number;
  $\mathbf{P}(\limsup A_n) \geq \limsup \mathbf{P}(A_n)$.

Stochastic Processes

- **Stochastic Process**: A collection of RVs indexed by time.
  - **View as a mapping**: $Z : T \times \Omega \to \mathbb{R}; (t, \omega) \mapsto Z_t(\omega)$.
  - **View as a sequence**: $\{Z_{t_1}, Z_{t_2}, \dots\}$, $t_1, t_2 \in T$ is a sequence of time-relavent RVs defined on prob. space $(\Omega, \mathcal{F}, \mathbf{P})$.
  - Once the sample path ω is fixed, values of the whole RV walk $\{Z_{t_1}(\omega), Z_{t_2}(\omega), \dots\}$ are fixed.
- **Bernoulli Process**: discrete-time discrete-value SP $\{X_i\}_{i\geq 1}$;
  $X_i \sim$ i.i.d. Bernoulli(p).
  - **Intuition**: keep tossing a coin and record if i-th round is a head as X_i.
  - **Independence**: seq. $X_{n+1}, X_{n+2}, \dots$ after n is a Bernoulli SP independent of $X_1, \dots, X_n$,
  - **Memoryless**: W_n = trials from n to next success $\sim$ Geo(p),
  - **Interarrival**: Y_k = time of k-th success, $T_1 = Y_1$, $T_k = Y_k - Y_{k-1}$; $T_i \sim$ i.i.d. Geo(p); $Y_k = \sum_{i=1}^k T_i$,
  - **Trials until** k**-th success**: $Y_k \sim$ Pascal(k, p),
    - $p_{Y_k}(t) = C_{t-1}^{k-1}p^k(1-p)^{t-k}$, $t = k, k+1, \dots$,
    - $\mathbf{E}[Y_k] = k/p$, $\mathrm{var}(Y_k) = k(1-p)/p^2$,
  - **Failures until** k**-th success**: $Z_k \sim$ NegBin(k, p),
    - $p_{Z_k}(t) = C_{t+k-1}^{k-1}p^k(1-p)^t$, $t = 0, 1, 2, \dots$,
    - $\mathbf{E}[Z_k] = k(1-p)/p$, $\mathrm{var}(Z_k) = k(1-p)/p^2$,
  - **Splitting**: keep w.p. $q \Rightarrow$ Bernoulli SP(pq),
  - **Merging Independent Bernoullis**:
    Bernoulli(p) OR Bernoulli$(q) \Rightarrow$ Bernoulli$(p + q - pq)$.
- **Poisson Process**: continuous-time discrete-value SP $\{N_t\}_{t\geq 0}$, $N_0 = 0$, $N_t \sim$ Poisson(λt); .
  - **Intuition**: count the number of arrivals w.p. $\lambda\delta$ for a small interval δ, from 0 to t as N_t.
  - **Homogeneity**: $\mathbf{P}(k, t) = \mathbf{P}(k$ arrivals in interval of duration $t)$
    $= \sum_{n=0}^\infty \mathbf{P}(N_{t+T} = n + k | N_T = n)\mathbf{P}(n, T)$
  - **Independent**: Numbers of arrivals in **disjoint** time intervals are independent.
  - **Small** δ: $P(0, \delta) = 1 - \lambda\delta + o(\delta)$, $P(1, \delta) = \lambda\delta + o(\delta)$, $P(k, \delta) = o(\delta)$, $k > 1$,
  - **Bernoulli Relation**: Select time infinitesimal δ, $n = t/\delta$, $p = \lambda\delta$; $\{X_i\}$ is a Bernoulli SP(p), $S_n = \sum_{k=1}^n X_i \sim$ Binomial RV$(n, p) \to N_t \sim$ Poisson RV (λt), $n \to \infty$,
  - **First Arrival**: $Y_1 = T_1 \sim$ Exp(λ); $F_{Y_1}(t) = 1 - e^{-\lambda t}$, $f_{Y_1}(t) = \lambda e^{-\lambda t}$;
  - **Interarrival**: Y_k = time of kth arrival, $T_1 = Y_1$, $T_k = Y_k - Y_{k-1}$; $T_i \sim$ i.i.d. Exp(λ), $Y_k = \sum_{i=1}^k T_i$,
  - k**-th arrival**: $Y_k \sim$ Erlang(λ, k),

    - $f_{Y_k}(y) = \lambda^k y^{k-1}e^{-\lambda y}/(k-1)!$, $y \geq 0$;
    - $\mathbf{E}[Y_k] = k/\lambda$, $\mathrm{var}(Y_k) = k/\lambda^2$,
  - **Splitting**: keep w.p. $p \Rightarrow$ Poisson(λp),
  - **Merging Independent Poissons**: Poisson(λ_1) + Poisson(λ_2) = Poisson$(\lambda_1 + \lambda_2)$;
    next arrival from SP 1 w.p. $\lambda_1/(\lambda_1 + \lambda_2)$.
- **Random Sum**: $Y = \sum_{i=1}^N X_i$, $Y = 0$ if $N = 0$; $\mathscr{M}_Y(s) = \mathscr{M}_N(\ln\mathscr{M}_X(s))$.
- **Random Sum Table**:

X_i	N	Y
Bernoulli(p)	Binomial(m, q)	Binomial(m, pq)
Bernoulli(p)	Poisson(λ)	Poisson(λp)
Bernoulli(p)	NegBin(r, q)	NegBin$(r, q/(q + p - pq))$
Geo(p)	Geo(q)	Geo(pq)
Exponential(λ)	Geo(q)	Exponential(λq)

Entropy

- **Discrete**: $H(X) = \mathbf{E}[-\ln p_X(X)] = -\sum_k p_X(x_k)\ln p_X(x_k)$.
- **Continuous**: $h(X) = \mathbf{E}[-\ln f_X(X)] = -\int f_X(x)\ln f_X(x)\,dx$.
- **Bernoulli**: $H(X) = -p\log_2 p - (1-p)\log_2(1-p)$; max 1 at $p = 1/2$, min 0 at $p = 0, 1$.
- **Max Entropy**:
  - finite $\{x_1, \dots, x_n\} \Rightarrow p_k = 1/n$,
  - $[a, b] \Rightarrow U(a, b)$,
  - $[0, \infty)$, $\mathbf{E}[X] = 1/\lambda \Rightarrow$ Exp(λ),
  - $\mathbb{R}$, fixed $\mu, \sigma^2 \Rightarrow N(\mu, \sigma^2)$.

Integration Formula

- $\int xe^{ax}\,dx = \frac{1}{a^2}(ax - 1)e^{ax}$, $\int x^2e^{ax}\,dx = \frac{a^2x^2 - 2ax + 2}{a^3}e^{ax}$, $\int x^3e^{ax}\,dx = \frac{a^3x^3 - 3a^2x^2 + 6ax - 6}{a^4}e^{ax}$, $\int x^ne^{ax}\,dx = \frac{1}{a}x^ne^{ax} - \frac{n}{a}\int x^{n-1}e^{ax}\,dx$.
- $\int_0^\infty x^ne^{-ax}\,dx = n!/a^{n+1}$.
- $\int_a^\infty x^ne^{-\lambda x}\,dx = e^{-\lambda a}\sum_{j=0}^n \frac{n!a^j}{j!\lambda^{n+1-j}}$.
- $\int_a^\infty \lambda e^{-\lambda x}\,dx = e^{-\lambda a}$; $\int_0^a \lambda e^{-\lambda x}\,dx = 1 - e^{-\lambda a}$.
- $\int_a^\infty \frac{\lambda^k x^{k-1}e^{-\lambda x}}{(k-1)!}\,dx = e^{-\lambda a}\sum_{j=0}^{k-1}\frac{(\lambda a)^j}{j!}$.
- $\int\sqrt{a^2 - x^2}\,dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\arcsin\frac{x}{a}$.
- $\int \ln x\,dx = x\ln x - x$, $\int x^n\ln x\,dx = \frac{1}{n+1}x^{n+1}\left(\ln x - \frac{1}{n+1}\right)$.
- $\int x\exp\left(-x^2/\sigma^2\right)\,dx = -\frac{\sigma^2}{2}\exp\left(-x^2/\sigma^2\right) + C$.
- $\int_{-\infty}^{+\infty}\exp\left(-(x - \lambda)^2/\mu^2\right)\,dx = |\mu|\sqrt{\pi}$.
- $\int_{-\infty}^{+\infty}x^2\exp\left(-x^2/\mu^2\right)\,dx = \frac{\mu^3}{2}\sqrt{\pi}$.